TREND TRACKERS

SALES TREND ANALYSIS

A Python-powered Business Intelligence Project

Data Analysis | Visualisation | Interactive Dashboard

Presented by: Trend Trackers
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Project Overview

Turning raw sales records into meaningful business insights

- Trend Trackers is a Sales Trend Analysis and Business Intelligence project.
- It analyses sales performance across months, regions, product categories and products.
- Python is used for data cleaning, calculations, analysis and visualisation.
- A Streamlit dashboard makes results interactive and easy to explore.

DATA	**Sales CSV**
ANALYSE	**Python + Pandas**
VISUALISE	**Charts + KPIs**
EXPLORE	**Interactive dashboard**

Problem Statement & Objectives

THE CHALLENGE

Sales records can be difficult to interpret when stored in rows and columns. Businesses need clear views of trends and performance to support decisions.

PROJECT OBJECTIVES

- Calculate total sales, cost, profit and profit margin.
- Track monthly sales movement.
- Compare regions, categories and products.
- Present insights in an interactive dashboard.
- Allow filtered data to be exported.

Tools & Technologies

Core technologies used in the project

Python

Main programming language

Pandas

Data cleaning and aggregation

Matplotlib

Static charts from analysis.py

Plotly

Interactive dashboard charts

Streamlit

Dashboard interface

CSV

Input sales dataset

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Project Methodology

End-to-end workflow from data to insight

01

Load

Read sales_data.csv

02

Clean

Convert data types; remove invalid rows

03

Calculate

Compute Sales, Cost and Profit

04

Summarise

Group by month, category, region and product

05

Visualise

Create charts and KPI summaries

06

Dashboard

Explore results with Streamlit filters

Key Performance Indicators

Illustrative results from the included synthetic demo dataset

TOTAL SALES	TOTAL PROFIT	UNITS SOLD	PROFIT MARGIN
$13,385	$4,200	53	31.4%

- Sales measures revenue generated from sold units.
- Profit is sales revenue minus product cost.
- Profit margin expresses profit as a percentage of sales.

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Visualisation 1: Monthly Sales Trend

Time-based view of sales performance across the sample period

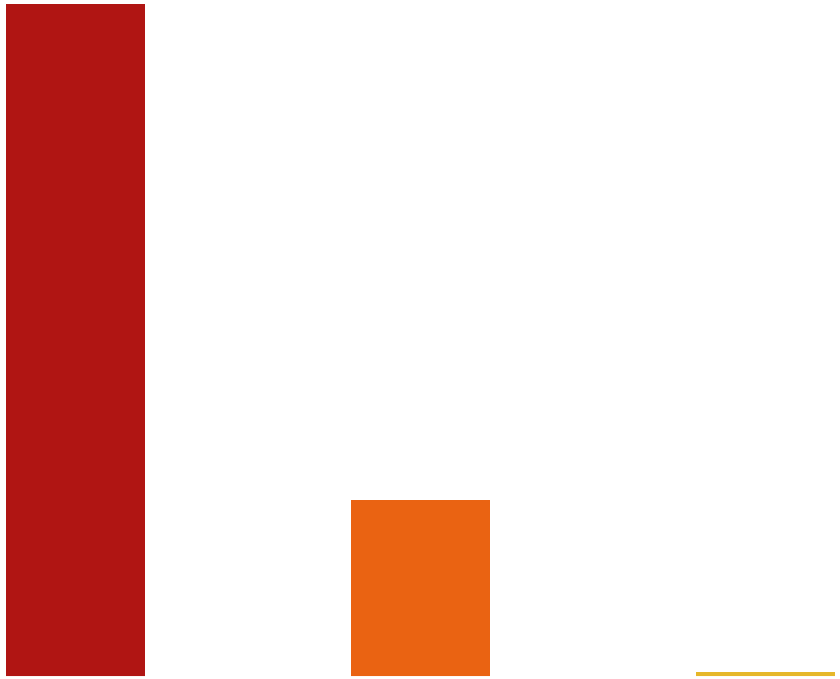


How to interpret

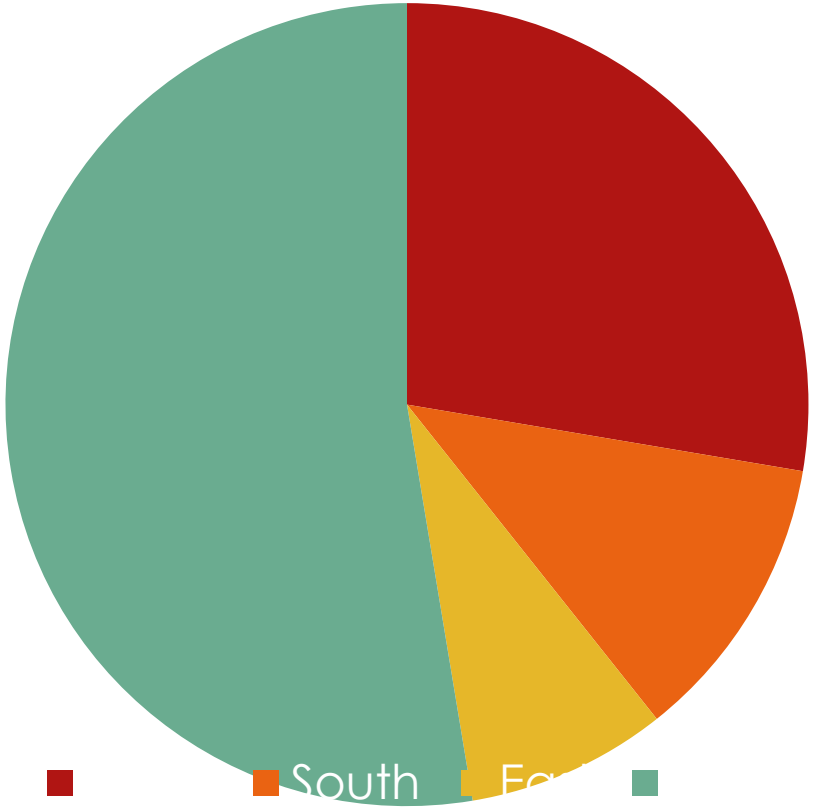
- X-axis shows the month.
- Y-axis shows sales value.
- The line makes month-to-month changes easy to compare.

Visualisation 2: Category & Region Performance

Sales by Category

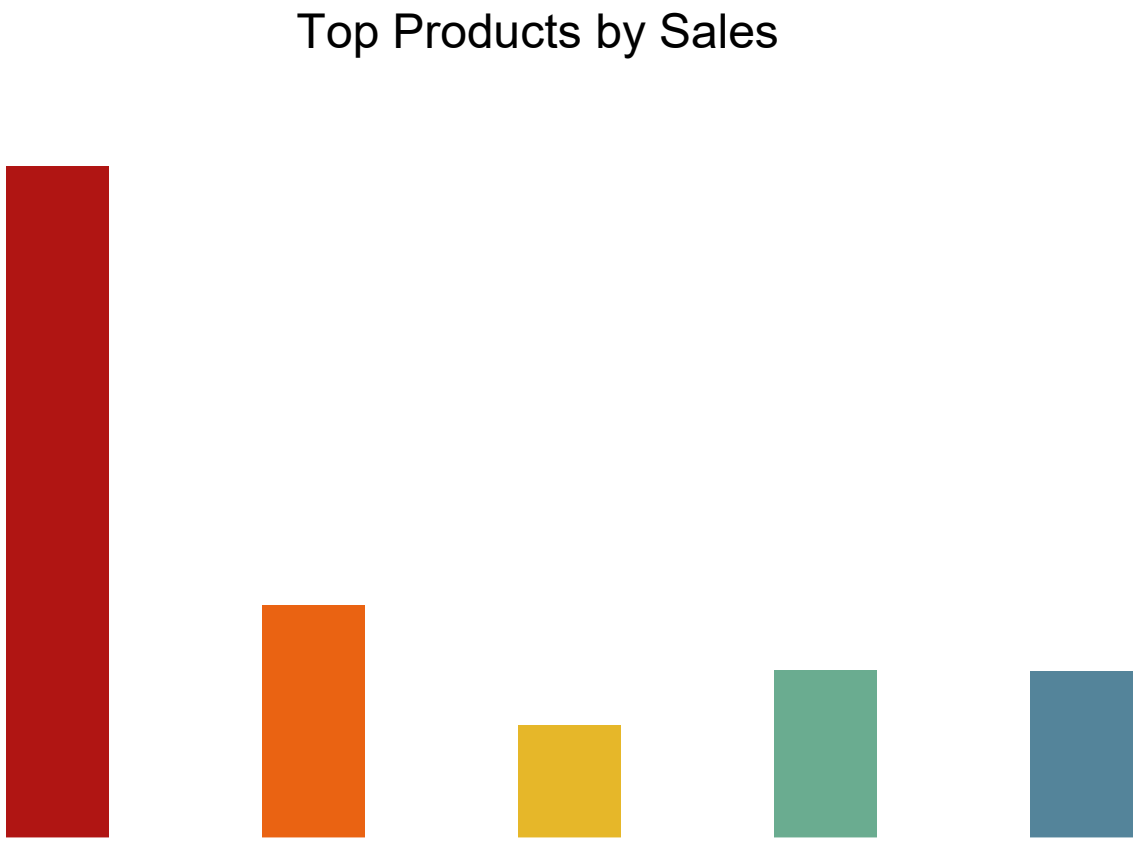


Sales Mix by Region



Visualisation 3: Top Products

Product-level sales comparison

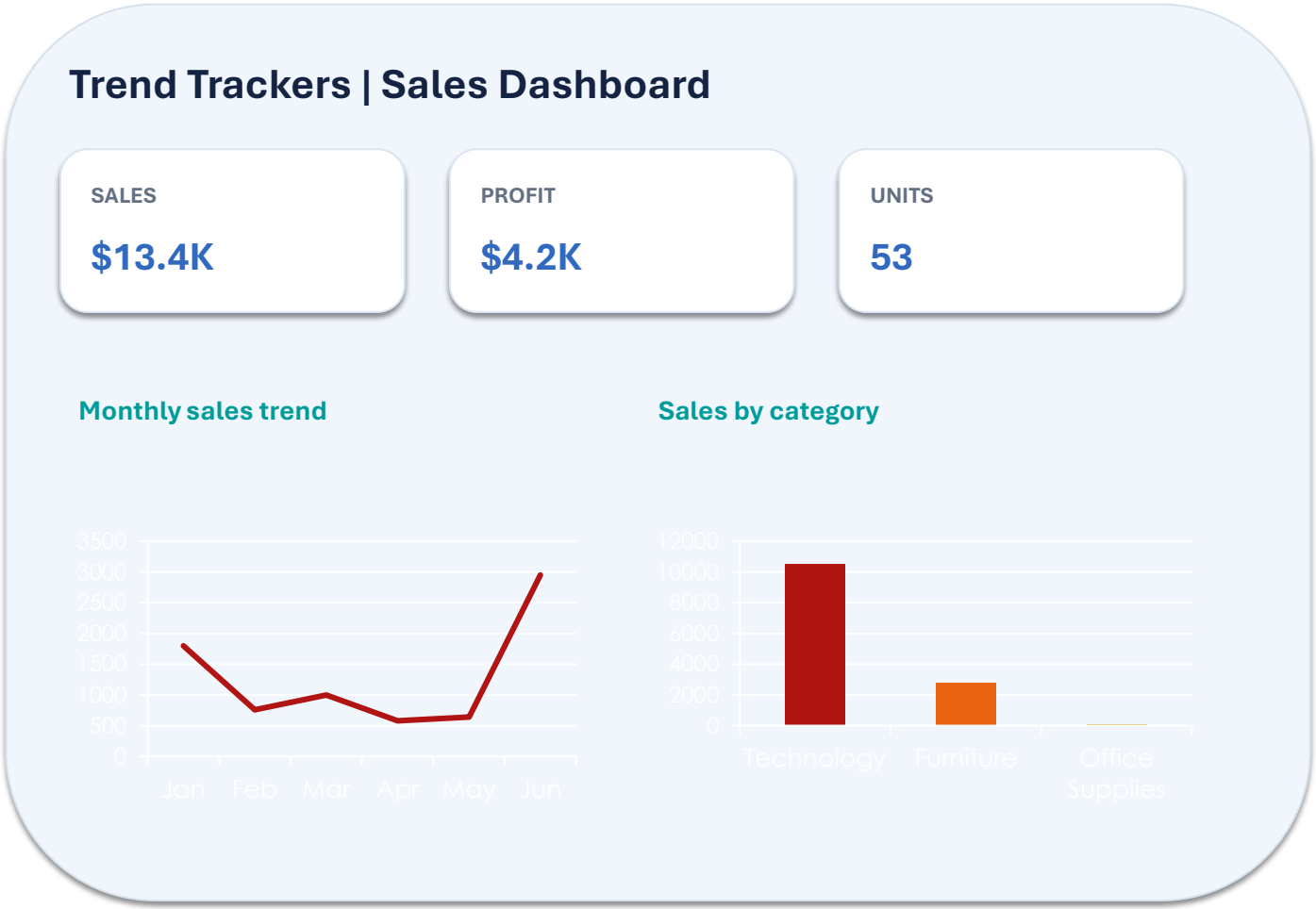


Business questions

- Which products generate the highest sales?
- Which products may need more inventory or marketing attention?
- How can product insights support planning?

Interactive Dashboard

Streamlit dashboard capabilities



Dashboard features

- Filters for year, region and category.
- Interactive charts respond to filters.
- View filtered sales records in a table.
- Download filtered records as CSV.

How to Run the Project

Setup and execution steps for analysis.py and the Streamlit dashboard

A. Run analysis.py

1. Open the project folder in VS Code.
2. Choose Terminal → New Terminal.
3. Install: python -m pip install pandas matplotlib
4. Run: python analysis.py
5. Charts and summary CSV files are saved in outputs/.

B. Run dashboard.py

1. Install: python -m pip install streamlit pandas plotly
2. Keep dashboard.py beside the data folder, or use its correct path.
3. Run: python -m streamlit run dashboard.py
4. If in src/: python -m streamlit run src/dashboard.py
5. Open the local URL shown in Terminal (usually localhost:8501).

Before running: confirm data/sales_data.csv exists. The starter analysis script can generate a demo CSV if it is missing.

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Project Outputs & Key Takeaways

- Cleaned sales dataset with calculated Sales, Cost and Profit fields.
- KPI summary with total sales, total cost, total profit, units sold and profit margin.
- Monthly, category, region and product summary CSV files.
- Static charts from analysis.py and interactive charts in Streamlit.
- Dashboard filters and CSV download support further exploration.

PROJECT VALUE

The project converts raw transaction records into visual summaries that make sales performance easier to monitor and discuss.

Limitations & Future Enhancements

CURRENT SCOPE

- Demonstration uses synthetic sample data.
- Results depend on input data quality.
- Dashboard is intended for local demonstration.

FUTURE SCOPE

- Connect to a live SQL database.
- Add sales forecasting and seasonality analysis.
- Add customer and salesperson performance views.
- Deploy dashboard for wider access.

THANK YOU

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Sales Trend Analysis